Paschen Curve and Gas Discharge

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The breakdown voltage of air was measured as a function of electrode gap d (0.2–5.5 mm) at 4, 6 and 8 hPa in a parallel-plate discharge cell, to test whether it depends only on the product pd of pressure and gap, as Townsend's theory predicts, and to extract the second Townsend coefficient γ. At 8 hPa the curve has a minimum of (416.8 ± 0.5) V at $(pd)_{\min} = (0.764 \pm 0.015)$ Pa·m. Using the manual's gas constants, the minimum voltage gives $\gamma = (1.846 \pm 0.013) \times 10^{-3}$ and the minimum position gives $(3.6 \pm 0.4) \times 10^{-3}$. The factor-of-two difference arises because the measured reduced field at the minimum, (545 ± 11) V/(Pa·m), exceeds the tabulated constant B by 12%; it limits γ far more than the measurement scatter does. The data do not obey pd scaling: the 6 hPa minimum lies at about three times the 8 hPa $(pd)_{\min}$ at nearly the same voltage, and the 4 hPa voltages exceed the 8 hPa voltages at equal pd by up to 263 V. Neither a change in γ, a constant pressure or gap offset, nor streamer breakdown accounts for this. The Townsend curve describes the data only near the 8 hPa minimum, and breakdown at gaps too small for Townsend breakdown indicates that the discharge did not stay within the uniform gap.

1. INTRODUCTION

At room temperature, air is an excellent insulator. The few free electrons produced by ambient radioactivity or short-wavelength radiation carry negligible current [1]. If, however, the electric field between two electrodes is strong enough, each free electron can ionise a neutral molecule on its way to the anode, and the liberated electron can ionise another, producing an exponentially growing cascade. Once the ions created in this cascade release enough secondary electrons from the cathode to restart the process, the discharge becomes self-sustaining: the gas conducts and a visible glow discharge appears. The voltage at which this occurs is the breakdown voltage U_{br}.

Paschen showed in 1889 that the breakdown voltage depends not on the pressure p and electrode separation d individually, but only on their product pd [2]. The number of collisions an electron undergoes while crossing the gap is proportional to the number of gas molecules per unit electrode area, which scales with pd. The curve of U_{br} against pd (the Paschen curve) is U-shaped. On the left branch, the gap is too short for enough ionising collisions; on the right, collisions are so frequent that electrons lose their energy to excitation and elastic scattering between collisions, and a higher voltage is needed to compensate.

This experiment measures the breakdown voltage of air at three pressures, tests whether the data depend only on pd, and extracts the second Townsend coefficient γ from the minimum of the curve.

2. THEORY

2.1 Electron Avalanche and the First Townsend Coefficient

Consider a free electron released at the cathode of a parallel-plate gap filled with gas at pressure p. The voltage U sets up a uniform field $E = U/d$ that accelerates the electron toward the anode. If the energy gained between collisions exceeds the ionisation threshold, a collision frees a second electron, and both continue to ionise. The resulting cascade is the electron avalanche (Fig. 1).

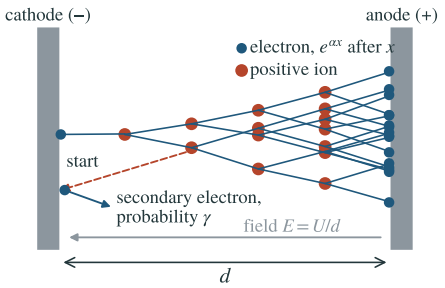


FIG. 1. Townsend discharge between parallel electrodes. An electron leaving the cathode ionises gas molecules as it drifts to the anode, so the number of electrons grows as $e^{\alpha x}$ (blue) and each ionisation leaves a positive ion (red). The ions drift back to the cathode, where each releases a secondary electron with probability γ, starting a new avalanche.

Townsend's first ionisation coefficient α is the number of ionising collisions per unit length of drift. If n electrons enter a slab of thickness dx, $n\alpha dx$ new electrons emerge, so the number grows exponentially [1]:

$$n(x) = n_0 e^{\alpha x} \tag{1}$$

A single electron starting at the cathode therefore produces $e^{\alpha d} - 1$ ion–electron pairs by the time it reaches the anode. The coefficient α depends on the field per unit pressure. For E/p below about 1140 V/(Pa·m) it is described by [1–3]

$$\alpha = Ape^{-Bp/E} = Ape^{-\frac{Bpd}{U}} \tag{2}$$

where A and B are gas-specific constants. This form was established for nitrogen; the manual assumes that it holds for air, which is 78% nitrogen, and lists $A = 20\ (\mathrm{Pa \cdot m})^{-1}$ and $B = 487\ \mathrm{V/(Pa \cdot m)}$ for air [1].

2.2 Secondary Emission and the Breakdown Criterion

An avalanche on its own is not breakdown. The positive ions drift back to the cathode, and a fraction of them release secondary electrons on impact (the Auger effect; Fig. 1). The probability per ion is the second Townsend coefficient γ, which depends on the gas, the cathode material and its surface condition [1].

The discharge becomes self-sustaining when a single avalanche produces enough ions to regenerate at least one starting electron. Since each of the $e^{\alpha d} - 1$ ions has probability γ of releasing an electron, the Townsend breakdown criterion is [1]

$$\gamma\left(e^{\alpha d} - 1\right) = 1. \tag{3}$$

2.3 The Paschen Law and Its Minimum

Substituting Eq. (2) into Eq. (3) and defining $C = \ln(1/\gamma + 1)$, the breakdown condition becomes $\alpha d = C$, that is, $Apde^{-(Bpd)/U} = C$. Taking logarithms and solving for U gives Paschen's law [1]:

$$U_{\mathrm{br}} = \frac{Bpd}{\ln(Apd/C)}. \tag{4}$$

U_{br} depends only on pd and the constants A, B and C, the last encoding the cathode surface through γ. Two gaps with the same pd should break down at the same voltage however p and d are chosen. A finite U_{br} exists only for $Apd > C$; at smaller pd no voltage satisfies Eq. (3), because even full ionisation efficiency ($\alpha = Ap$) gives too few ionising collisions across the gap.

Setting the derivative of Eq. (4) with respect to pd to zero gives the coordinates of the minimum [1]:

$$(pd)_{\mathrm{min}} = \frac{e}{A}\ln\!\left(\frac{1}{\gamma} + 1\right) \tag{5}$$

$$U_{\mathrm{min}} = \frac{eB}{A}\ln\!\left(\frac{1}{\gamma} + 1\right) \tag{6}$$

where e is Euler's number. Either coordinate gives γ when A and B are known. Their ratio, the reduced field at the minimum, $(E/p)_{\mathrm{min}} = U_{\mathrm{min}}/(pd)_{\mathrm{min}} = B$, is independent of γ and tests whether the measured minimum is consistent with the constants.

3. METHODS

3.1 Apparatus

The setup followed the manual [1,4]. Fig. 2 shows its parts. Two parallel plate electrodes sat in a chamber evacuated by a vacuum pump; the pressure was set with a fine control valve and read from a digital pressure meter. One electrode was mounted on a micrometer screw, which set the gap d between 0 and 6 mm. The plasma physics operating unit supplied the electrode voltage. A digital multimeter on its 1000 V DC range, in peak-hold mode, recorded the highest voltage reached; at breakdown the current rises and the electrode voltage falls, so the held value is U_{br} [1]. The accuracy of the pressure meter and the multimeter, and the resolution of the micrometer, were not recorded.

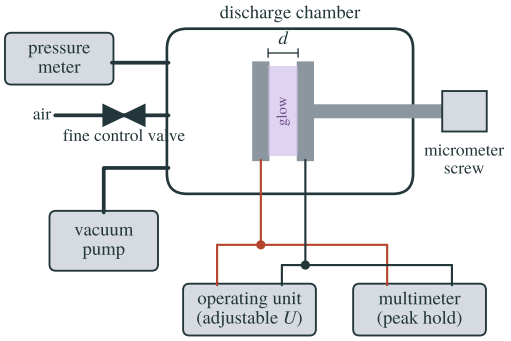


FIG. 2. Apparatus. The discharge chamber holds two parallel electrodes, one moved by a micrometer screw to set the gap d. A vacuum pump evacuates the chamber, air is admitted through a fine control valve, and a pressure meter reads the pressure. The operating unit applies the voltage U across the electrodes, and a multimeter in peak-hold mode records the highest voltage reached before breakdown.

3.2 Procedure

The manual prescribes 2, 4 and 6 hPa [1]. At 2 hPa breakdown could not be reached at the operating unit's maximum voltage except at the widest gaps, and 4, 6 and 8 hPa were used instead so that at least one pressure would show a complete minimum. The gap was zeroed by closing the electrodes and then set to the 23 values of the manual's protocol from 0.2 to 5.0 mm, plus 5.5 mm [1]. At each gap the voltage was raised slowly until a glow discharge appeared, more slowly than 1 V/s near breakdown at 6 and 8 hPa as the manual requires above 5 hPa, then lowered until the glow vanished, and the peak-hold value was recorded. Three trials were made per gap.

Fig. 3 shows the data sheet. At 4 hPa no breakdown voltage was recorded at 2.8 mm, and data were taken from 3.2 to 5.5 mm. An earlier 4 hPa series at 0.2–1.0 mm was discarded and is excluded from the analysis.

3.3 Analysis

Each point in the figures is the mean of the three trials, with the sample standard deviation σ as its uncertainty. The 8 hPa minimum was located by a least-squares parabola through the 24 individual trials at $d = 0.6$–1.4 mm. The uncertainties of d_{min} and U_{min} were propagated from the fit covariance, which is scaled by the residual scatter, and the effect of the fit window was assessed by repeating the fit over five other windows between 0.4 and 1.8 mm.

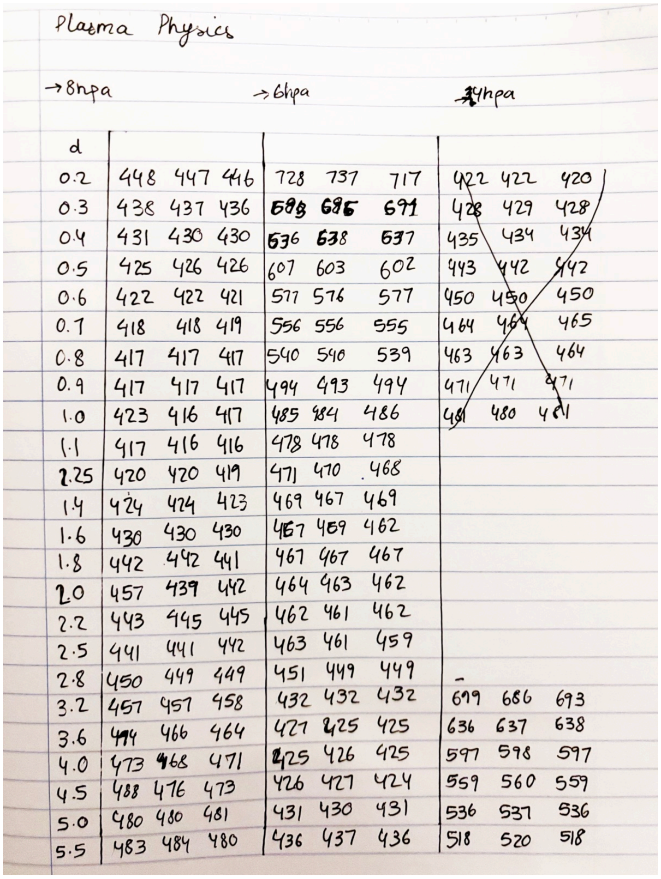


FIG. 3. Data sheet, with three breakdown-voltage trials at each gap for 8, 6 and 4 hPa. The struck-out 4 hPa series at $d = 0.2$–1.0 mm was discarded, and the dash at 2.8 mm marks a gap with no recorded value.

4. RESULTS

4.1 Breakdown Voltage versus Gap

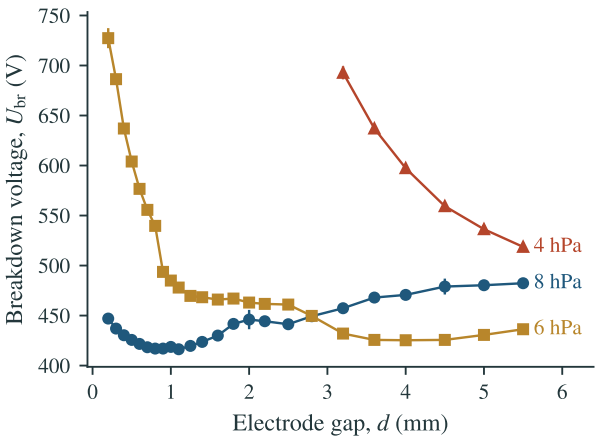


FIG. 4. Breakdown voltage U_{br} versus electrode gap d at 4, 6 and 8 hPa. Each point is the mean of three trials; error bars show $\pm\sigma$ and are mostly smaller than the markers.

The minimum moved to larger gaps as the pressure fell, but much further than pd scaling allows (Table I, Fig. 4). At 8 hPa the breakdown voltage fell from 447 V at $d = 0.2$ mm to 416–417 V between 0.8 and 1.1 mm, rose to 442 V at 1.8 mm, stayed at 441–446 V up to 2.5 mm, and rose to 482 V at 5.5 mm.

TABLE I. Breakdown voltage $\overline{U} \pm \sigma$ (V), the mean and sample standard deviation of three trials (Fig. 3), at each gap and pressure. A dash marks a gap without data.

d (mm)	8 hPa	6 hPa	4 hPa
0.20	447 ± 1.0	727 ± 10.0	–
0.30	437 ± 1.0	686 ± 4.2	–
0.40	430 ± 0.6	637 ± 1.0	–
0.50	426 ± 0.6	604 ± 2.6	–
0.60	422 ± 0.6	577 ± 0.6	–
0.70	418 ± 0.6	556 ± 0.6	–
0.80	417 ± 0.0	540 ± 0.6	–
0.90	417 ± 0.0	494 ± 0.6	–
1.00	419 ± 3.8	485 ± 1.0	–
1.10	416 ± 0.6	478 ± 0.0	–
1.25	420 ± 0.6	470 ± 1.5	–
1.40	424 ± 0.6	468 ± 1.2	–
1.60	430 ± 0.0	466 ± 3.6	–
1.80	442 ± 0.6	467 ± 0.0	–
2.00	446 ± 9.6	463 ± 1.0	–
2.20	444 ± 1.2	462 ± 0.6	–
2.50	441 ± 0.6	461 ± 2.0	–
2.80	449 ± 0.6	450 ± 1.2	–
3.20	457 ± 0.6	432 ± 0.0	693 ± 6.5
3.60	468 ± 5.3	426 ± 1.2	637 ± 1.0
4.00	471 ± 2.5	425 ± 0.6	597 ± 0.6
4.50	479 ± 7.9	426 ± 1.5	559 ± 0.6
5.00	480 ± 0.6	431 ± 0.6	536 ± 0.6
5.50	482 ± 2.1	436 ± 0.6	519 ± 1.2

Seventeen of the 24 gaps had $\sigma \le 1$ V. At 6 hPa the curve fell steeply from 727 V at 0.2 mm, including a 46 V step between 0.8 and 0.9 mm, levelled at 461–470 V between 1.25 and 2.5 mm, and fell again to 425–426 V at 3.6–4.5 mm before rising to 436 V at 5.5 mm. At 4 hPa the voltage fell from 693 V at 3.2 mm to 519 V at 5.5 mm without reaching a minimum. If U_{br} depended only on pd, the 8 hPa minimum near 0.96 mm (Sec. 5.1) would appear at 1.3 mm at 6 hPa and 1.9 mm at 4 hPa.

4.2 Test of pd Scaling

The three data sets do not collapse onto one curve when plotted against pd (Fig. 5). For $pd = 2.2$–3.3 Pa·m the 6 hPa voltages were 21–38 V below the 8 hPa voltages at the same pd, while the 4 hPa voltages were 71–263 V above them. The 4 hPa data were still falling at $pd = 2.2$ Pa·m, where the 8 hPa curve had been rising for more than 1 Pa·m.

The Townsend curve of Eq. (4) described only the region around the 8 hPa minimum. With γ chosen to match U_{min}, the 8 hPa data agreed with it within 7 V for $pd = 0.72$–1.12 Pa·m ($d = 0.9$–1.4 mm). On the right branch the data fell below the curve by 85 V at $pd = 2.0$ Pa·m and by 330 V at 4.4 Pa·m, where the curve predicts 812 V against the measured 482 V. On the left branch the curve rises far more steeply than the data and predicts no breakdown at all below $pd = 0.32$ Pa·m, yet breakdown occurred at all six such gaps (8 hPa: 0.2 and 0.3 mm; 6 hPa: 0.2–0.5 mm).

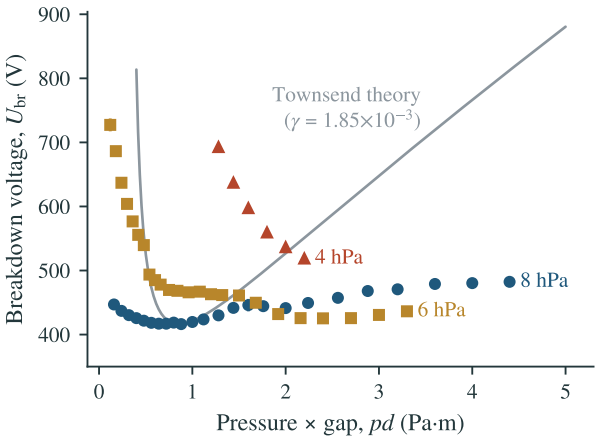


FIG. 5. Breakdown voltage versus pd for all three pressures, with Eq. (4) (grey) for the manual's A and B and $\gamma = 1.85 \times 10^{-3}$, the value that reproduces the measured 8 hPa minimum voltage. Agreement at the minimum is therefore built in; the test is the shape and the collapse of the three data sets. The curve is undefined below $pd = C/A = 0.32$ Pa·m.

5. ANALYSIS

5.1 Second Townsend Coefficient from the 8 hPa Minimum

Only the 8 hPa series resolves a minimum, so γ was extracted from it. The parabolic fit gave $d_{\min} = (0.955 \pm 0.019)$ mm and $U_{\min} = (416.8 \pm 0.5)$ V; across the six fit windows $d_{\min}$ ranged from 0.93 to 0.99 mm and $U_{\min}$ from 416.6 to 416.9 V. The minimum is $(pd)_{\min} = (0.764 \pm 0.015)$ Pa·m.

From $U_{\min}$ and Eq. (6), $C = U_{\min}A/(eB) = 6.297 \pm 0.007$ and

$$\gamma = \frac{1}{e^C - 1} = (1.846 \pm 0.013) \times 10^{-3}.$$

The statistical uncertainty is small because $|\delta\gamma/\gamma| \approx \delta C = \delta U_{\min}/(66\ \text{V})$: each volt of error in $U_{\min}$, including any multimeter calibration offset, changes γ by 1.5%. The result depends much more strongly on the assumed constants: a 1% change in B changes γ by 6.4%.

From $(pd)_{\min}$ and Eq. (5), which uses A but not B, $C = 5.62 \pm 0.11$ and $\gamma = (3.6 \pm 0.4) \times 10^{-3}$, with 3.0 to 4.3×10^{-3} across the fit windows. The two routes disagree by a factor of two because the reduced field at the minimum, $U_{\min}/(pd)_{\min} = (545 \pm 11)$ V/(Pa·m), is 12% above the manual's $B = 487$ V/(Pa·m). The manual's constants therefore do not describe this minimum exactly, and γ is fixed only to within this factor. A further test confirms that the measured minimum is broader than the Townsend form allows: fitting Eq. (4) to the trials at $d = 0.7$–1.25 mm with B and C both free leaves an rms residual of 6.5 V, against a pooled trial scatter of 1.5 V in the same region, and wider windows give 8–15 V.

The 6 hPa minimum gives no usable γ. Its reduced field, $U_{\min}/(pd)_{\min} = 425$ V /2.4 Pa·m = 177 V/(Pa·m), is a third of B, so the two routes would give values of γ that differ by orders of magnitude.

5.2 Failure of pd Scaling

The 6 hPa minimum lies at about three times the 8 hPa $(pd)_{\min}$, and no correction within the Townsend description accounts for this. The 6 hPa voltages at 3.6–4.5 mm agree within 1 V, so its minimum lies at $(pd)_{\min} = 2.2$–2.7 Pa·m, 2.8 to 3.5 times the 8 hPa value. Its minimum voltage, 425.3 V, is only 2% above the 8 hPa value. In Eq. (5) and Eq. (6) both coordinates are proportional to C, so a change in γ caused by a change in the electrode surface between series would move $U_{\min}$ and $(pd)_{\min}$ by the same factor; it cannot move one by a factor of three and the other by 2%.

Instrumental offsets are also excluded. A constant pressure-meter offset large enough to bring the two minima to the same pd would have to be 5.2 hPa, which would make the true pressure of the 4 hPa series negative. A micrometer zero error would have to be 7.6 mm, more than the 6 mm range of the screw.

6. DISCUSSION

6.1 Validity of the Townsend Framework

Streamer breakdown, in which the space charge of a single avalanche distorts the field enough to propagate a discharge, does not explain the low right branch. At Townsend breakdown each avalanche contains $e^{\alpha d} = 1 + 1/\gamma \approx 280$–$540$ electrons for the two values of γ above, whereas streamer formation needs about 10^8 [2].

The measurements constrain, but do not identify, the cause of the deviations. Three observations bear on it. First, breakdown at $pd < C/A$ is impossible in a uniform gap of length d, so at the smallest gaps the discharge must have followed a longer path, for example around the electrode edges, or the field cannot have been uniform. These left-branch points also have E/p above the 1140 V/(Pa·m) validity limit of Eq. (2) (at $d \leq 0.4$ mm at 8 hPa and $d \leq 0.7$ mm at 6 hPa). Second, the measured minimum is broader than any Townsend curve, and the right branch rises far more slowly. This is consistent with an $\alpha(E/p)$ that differs from Eq. (2) with the manual's constants, as expected if the gas was not clean air; the manual notes that contaminants change the results significantly, and backstreaming pump oil is one possible source [1]. Third, any fixed gas composition preserves pd scaling, so the different curves at 4, 6 and 8 hPa require something that changed between series, such as the gas composition, the electrode surface or the discharge path. The present data cannot distinguish these.

6.2 Trial Scatter and Uncertainty

Trial-to-trial scatter was small compared with the deviations from theory. Forty of the 54 gaps had $\sigma \leq 1.2$ V. The largest spreads occurred on the steep 6 hPa left branch ($\sigma = 10$ V at 0.2 mm) and at isolated 8 hPa gaps. In five of the six 8 hPa gaps with $\sigma > 2$ V (1.0, 2.0, 3.6, 4.0 and 4.5 mm) the first trial was the highest, by 4–17 V above the mean of the other two. This is consistent with an overshoot in the first voltage ramp at a new gap, which the peak-hold reading captures, or with the statistical delay before an initial electron starts the breakdown, which the manual's slow ramp is intended to avoid [1].

The uncertainty in γ is dominated by the constants A and B, not by the measurements. The statistical uncertainties of the two routes are 0.7% and 11%, but the routes differ by a factor of two. The multimeter and pressure-meter calibrations were not assessed; a 1% calibration error in either would change γ by about 6%, through $U_{\min}$ or $(pd)_{\min}$ respectively, far less than the disagreement between the routes.

7. CONCLUSION

The 8 hPa series resolved a Paschen minimum at $d_{\min} = (0.955 \pm 0.019)$ mm, $(pd)_{\min} = (0.764 \pm 0.015)$ Pa·m and $U_{\min} = (416.8 \pm 0.5)$ V. With the manual's A and B, the minimum voltage gives $\gamma = (1.846 \pm 0.013) \times 10^{-3}$ and the minimum position gives $\gamma = (3.6 \pm 0.4) \times 10^{-3}$. The two disagree because the measured reduced field at the minimum, (545 ± 11) V/(Pa·m), exceeds the manual's B by 12%, so γ is known only to within a factor of two.

The breakdown voltage did not depend on pd alone. The 6 hPa minimum lay at about three times the 8 hPa $(pd)_{\min}$ at nearly the same voltage, and the 4 hPa voltages exceeded the 8 hPa values at equal pd by up to 263 V. Neither a change in γ nor a constant pressure or gap offset accounts for this. The Townsend curve described the data only near the 8 hPa minimum. Breakdown also occurred at pd below the Townsend threshold, and streamer breakdown is excluded.

Four improvements follow from these limitations. Repeating the 8 hPa series after the other pressures would show whether the cell changed between series, and cleaning the electrodes with alcohol beforehand, as the manual suggests [1], would remove one candidate cause. Observing where the glow forms at the smallest gaps would test whether the discharge leaves the uniform gap. Calibrating the pressure meter and multimeter against reference instruments would bound the calibration contributions that were not assessed here. Finally, gap steps of 0.05 mm around the minimum would reduce the dependence of $d_{\min}$ on the fit window, which currently spans 0.93–0.99 mm and limits the $(pd)_{\min}$ route.

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